# Practice workbook · State what antibiotics treat and explain how resistance develops

What antibiotics are and where they come from, why they fail against viruses and fungi, and selection of resistant strains such as MRSA and its consequences.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIBIOTICS-AND-RESISTANCE

# 1. Question 1 · part b · 2 marks

November 2020 Paper 2

Worked-example retry · Attempt part b only.

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**1** (a) Different types of organisms cause diseases.

Diseases are transmitted in different ways.

The table lists three diseases.

Complete the table by giving the type of organism that causes each disease and the method of transmission for each disease.

**(6)**

| Name of disease | Type of organism | Method of transmission |
|---|---|---|
| malaria | | |
| poliomyelitis | | |
| typhoid | | |

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(b) Explain why it is not possible to treat athlete's foot using antibiotics.

**(2)**

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**(Total for Question 1 = 8 marks)**

## 1. Question 1 · part b · 2 marks · Working and answer

Working and answer

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# 2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

**6** *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

| Age group in years | Number of deaths in one year per million people | | | |
|---|---|---|---|---|
| | Males | | Females | |
| | *S. aureus* | MRSA | *S. aureus* | MRSA |
| under 45 | 1.8 | 0.5 | 1.4 | 0.4 |
| 45–54 | 6.0 | 2.5 | 4.3 | 2.4 |
| 55–64 | 15.7 | 8.2 | 8.1 | 4.2 |
| 65–74 | 41.2 | 26.1 | 21.9 | 12.5 |
| 75–84 | 128.6 | 86.8 | 81.1 | 51.6 |
| 85 and over | 416.6 | 299.9 | 200.3 | 146.4 |

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

**(2)**

1 .....

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## 2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

**(2)**

ratio = .....

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

**(2)**

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number = .....

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

**(2)**

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## 2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

**(1)**

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

**(6)**

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**(Total for Question 6 = 15 marks)**

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## 2. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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## 3. Question 7 · part a.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

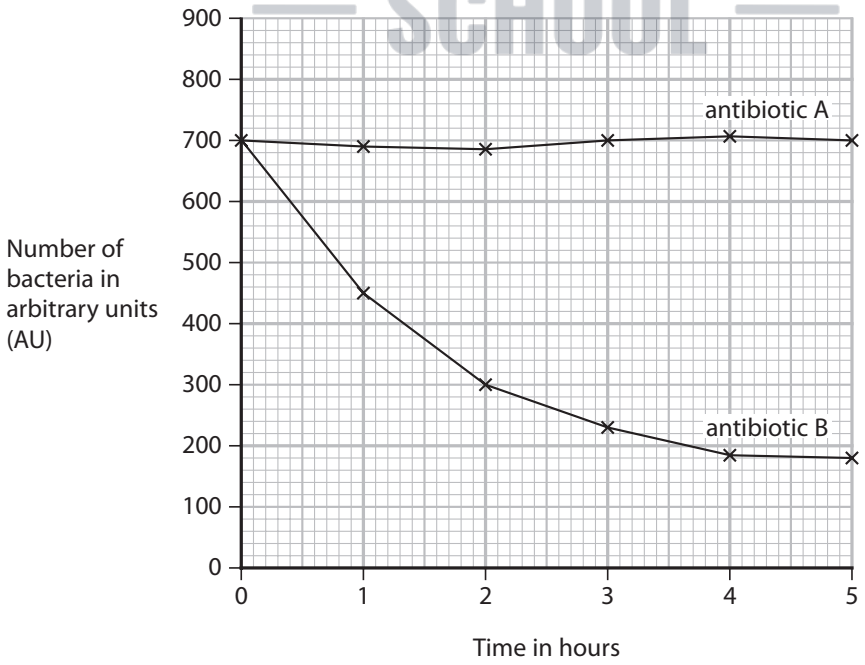
**7** Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.




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## 3. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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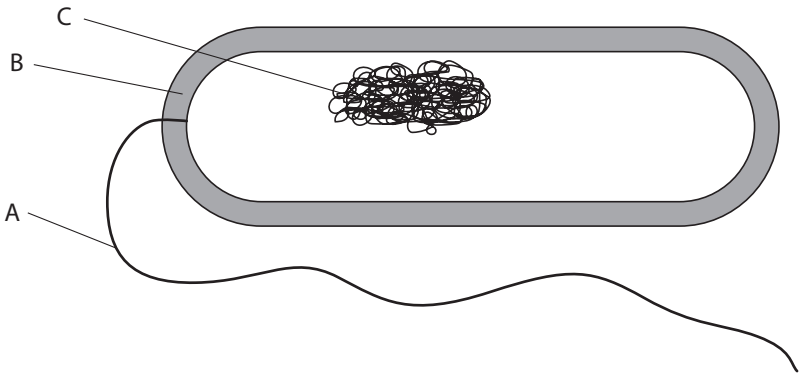
# 4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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**8** The diagram shows a bacterial cell.




(a) (i) Name the parts labelled A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

**(1)**

.....

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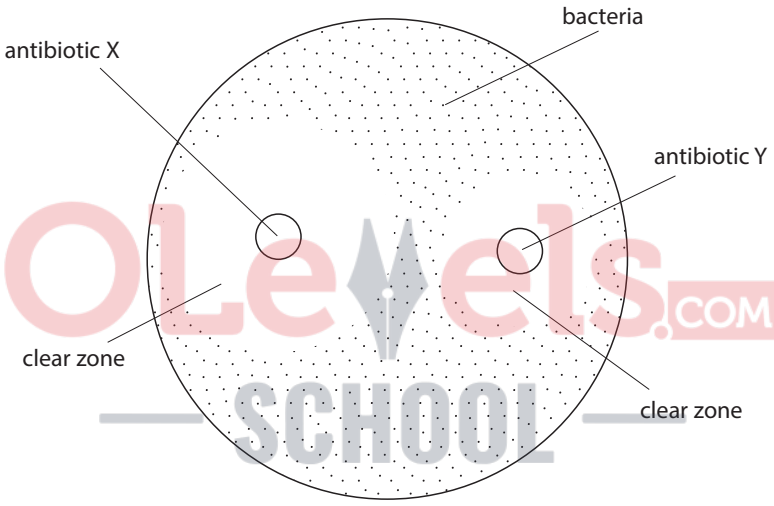
# 4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days




(i) Describe what is meant by the term **antibiotic**.

**(2)**

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## 4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

**(3)**

(iii) Explain a suitable control that could be used in this investigation.

**(3)**

**(Total for Question 8 = 12 marks)**

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## 4. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

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# 5. Question 4 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

**4** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

**(2)**

diarrhoea

malaria

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# 5. Question 4 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

**(1)**

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

**(2)**

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

**(2)**

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

**(3)**

number of people = .....

## 5. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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# 6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

**6** *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

| Age group in years | Number of deaths in one year per million people | | | |
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| 75–84 | 128.6 | 86.8 | 81.1 | 51.6 |
| 85 and over | 416.6 | 299.9 | 200.3 | 146.4 |

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

**(2)**

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# 6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

**(2)**

ratio = .....

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

**(2)**

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number = .....

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

**(2)**

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# 6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

**(1)**

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

**(6)**

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**(Total for Question 6 = 15 marks)**

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## 6. Question 6 · part a.iv · 2 marks · Working and answer

Working and answer

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# 7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

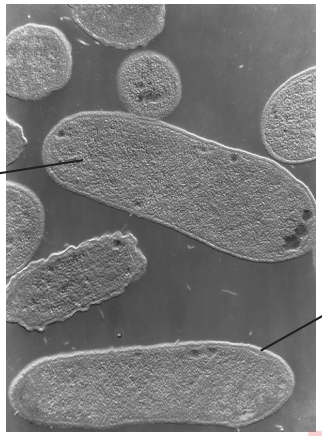
Further practice · Attempt part c.ii only.

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**4** Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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# 7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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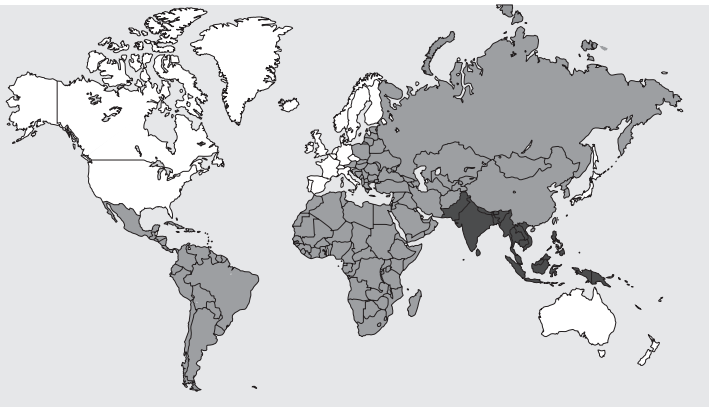
# 7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(c) The diagram shows the levels of typhoid across the world.




(i) Give reasons for the global distribution of typhoid.

**(2)**

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# 7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(ii) Some strains of bacteria that cause typhoid are resistant to antibiotics.

Discuss how resistance to antibiotics develops and the effect this has on the treatment of typhoid.

**(5)**

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**(Total for Question 4 = 14 marks)**

## 7. Question 4 · part c.ii · 5 marks · Working and answer

Working and answer

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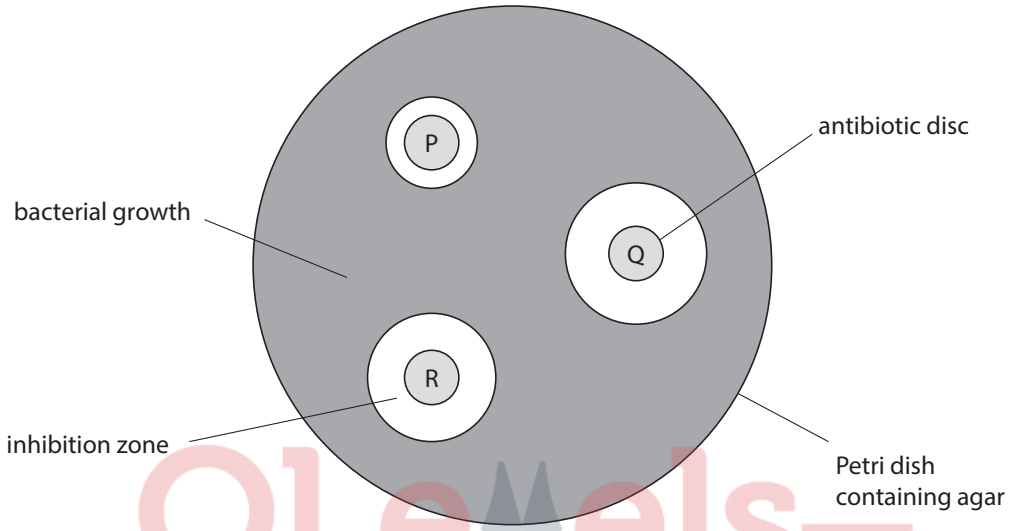
# 8. Question 5 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

**5** (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.




(i) Which is the independent variable in this investigation?

**(1)**

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

**(2)**

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2 .....

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P 7 4 6 3 0 A 0 1 4 2 4

# 8. Question 5 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

(iii) Explain which is the most effective antibiotic.

**(2)**

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

**(4)**

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

**(2)**

**(Total for Question 5 = 11 marks)**

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## 8. Question 5 · part a.iv · 4 marks · Working and answer

Working and answer

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# 9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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**8** (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

**(1)**

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

**(1)**

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

**(2)**

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# 9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

**(1)**

(v) Suggest what is meant by a specific immune response. (line 7)

**(2)**

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

**(2)**

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# 9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

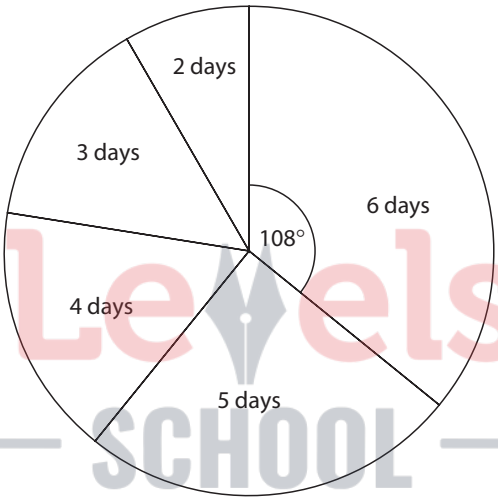
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.




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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

**(3)**

number of types of antibiotic = .....



P 7 4 6 3 2 A 0 2 3 2 4

*Turn over* ▶

## 9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

**(1)**

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

**(2)**

**(Total for Question 8 = 15 marks)**

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Working and answer